



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

**FINAL EXAMINATION
FEBRUARY 2023**

Course : EC3105 (C Programming)

Time : 8.00 am – 11.00 am
(3 hours)

Lecturer : Noor Zuhaili bt Md Yasin

Date : 25 May 2023

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of **5** printed pages
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1.
 - a. Define the UNIX Operating System. (5 MARKS)
 - b. How to view your username in Unix? Explain your answer. (4 MARKS)
 - c. What is a File in Unix? (4 MARKS)
 - d. List any **TWO (2)** basic UNIX commands for working with files. (2 MARKS)
 - e. List any **FIVE (5)** basic UNIX users commands. (5 MARKS)
 - f. Differentiate between multiuser and multitasking in UNIX operating system. (5 MARKS)

2.
 - a. Discuss any **TWO (2)** uses of C programming. (6 MARKS)
 - b. Compute the values of the following C expressions assuming that a, b and c are integer variables as declared below:

$$\text{int } a=10, b=12, c=2;$$
 - i. $x = a - b / 3 + c * 2 - 1$
 - ii. $y = a - b / (3 + c) * (2 - 1)$
 - iii. $z = a - (b / (3 + c) * 2) - 1$
 - c. List **FIVE (5)** rules of naming identifiers. (5 MARKS)
 - d. Write down a C expression corresponding to each of the following mathematical expressions: (4 MARKS)
 - i. $-c + (c^2 - 4ac) / 2a$
 - ii. $t/k - tk + 2/3s^2$
 - e. List any **FOUR (4)** examples of identifiers in C programming. (4 MARKS)

3. a. What will be the output of following program?

(3 MARKS)

```
#include <stdio.h>
enum birds {SPARROW, PEACOCK, PARROT};
enum animals {TIGER = 8, LION, RABBIT, ZEBRA};
int main()
{
    enum birds m = TIGER;
    int k;
    k = m;
    printf("%d\n", k);
    return 0;
}
```

- b. What is the difference between the following two C programming codes? Explain your answer.

(2 MARKS)

```
1. #include <stdio.h> //Program 1
2. int main()
3. {
4.     int d, a = 1, b = 2;
5.     d = a++ + ++b;
6.     printf("%d %d %d", d, a, b);
7. }
```

Program 1

```
1. #include <stdio.h> //Program 2
2. int main()
3. {
4.     int d, a = 1, b = 2;
5.     d = a++ ++b;
6.     printf("%d %d %d", d, a, b);
7. }
```

Program 2

c. What is the output of the following program?

(6 MARKS)

```
//Example 1
#include <stdio.h>
int main()
{
    int* pc, c;

    c = 22;
    printf("Address of c: %p\n", &c);
    printf("Value of c: %d\n\n", c); // 22

    pc = &c;
    printf("Address of pointer pc: %p\n", pc);
    printf("Content of pointer pc: %d\n\n", *pc); // 22

    c = 11;
    printf("Address of pointer pc: %p\n", pc);
    printf("Content of pointer pc: %d\n\n", *pc); // 11

    *pc = 2;
    printf("Address of c: %p\n", &c);
    printf("Value of c: %d\n\n", c); // 2
    return 0;
}
```

d. Identify the errors and their types in the following C Programming codes:

(8 MARKS)

```
i. #include<stdio.h>
main() {
    printf("Hello World")
}
```

```

ii. #include<stdio.h>
main() {
    int x = 52;
    int y = 0;
    printf("Div : %f", x/y);
}

```

```

iii. #include<stdio.h>
Main() {
    int x = 52;
    int y = 0;
    printf("Div : %f", x/y);
}

```

```

iv. #include<stdio.h>
main() {
    int i;
    for(i = 0; i<5; i++); {
        printf("Hello World");
    }
}

```

- e. The code below is the incomplete C code. Rewrite the following code by using the function to read a line of string and to display the string. **(6 MARKS)**

```

#include <stdio.h>
int main()
{
    char name[30];
    printf("Enter name: ");
    printf("Name: ");
    return 0;
}

```

4. a. Explain what is meant by looping/iteration statements in C programming. **(3 MARKS)**
- b. Discuss **THREE (3)** types of loops in C programming. **(6 MARKS)**
- c. By using suitable loop, write a program in C to calculate the sum of first n natural numbers. (Hint: Positive integers 1,2,3... are known as natural numbers). **(5 MARKS)**
- d. Write a program in C to display such a pattern for n number of rows using a number which will start with the number 1 and the first and a last number of each row will be 1. The pattern is as follows: **(6 MARKS)**
- ```
1
121
12321
```
- e. List any **FOUR (4)** major operations you can perform on files. **(4 MARKS)**
- f. Write a syntax for opening a file in standard input and output (I/O). **(1 MARK)**

**-END OF QUESTION PAPER-**